

Generative AI in Multimedia: Challenges and Opportunities for Academic and Industrial Impact

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Abstract

Generative AI has revolutionized multimedia, leading to groundbreaking developments in content creation, interactive experiences, and personalized media. This panel delves into the transformative potential of generative AI in academic and industrial sectors, exploring its future applications and connections to emerging techniques. Additionally, the panel will address newly identified opportunities and challenges from both technical and ethical perspectives, highlighting the importance of responsible AI development. Bringing together leading experts from universities, research institutions, and industry, this panel aims to foster discussion and debate among participants. We invite everyone to join and contribute to this critical and promising area of research in the multimedia community.

CCS Concepts

• Computing methodologies → Artificial intelligence.

Keywords

Generative AI, Large Foundation Model, Multimedia

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1 Introduction

Generative AI, encompassing technologies like Large Language Models (LLMs) and Diffusion Models, has significantly impacted the multimedia community. These models, characterized by their ability to generate high-quality, realistic content, have opened new avenues for research and innovation. In academia, generative AI fosters new research paradigms and interdisciplinary collaborations, while in industry, it drives the development of innovative products and services. Generative AI models, such as GPT-4, DALL-E, and Stable Diffusion, have been trained on vast datasets, often exceeding hundreds of terabytes of text, images, and videos. This scale of data enables these models to produce outputs that are not only coherent but also creatively novel. For instance, GPT-4, with its 175

billion parameters, demonstrates a remarkable ability to generate human-like text, while DALL-E 2 can create photorealistic images from textual descriptions, pushing the boundaries of what AI can achieve in multimedia content creation. Recently, Sora, a generative video model, has emerged as a groundbreaking innovation in this space. Sora can generate high-quality, realistic video content from textual descriptions, enabling new possibilities for video production, entertainment, and virtual reality. This model represents a significant advancement, providing tools for creating dynamic and immersive video experiences. The economic impact of generative AI is equally profound. According to a report by PwC, AI could contribute up to \$15.7 trillion to the global economy by 2030, with a significant portion driven by advancements in generative AI. In the entertainment industry alone, the global market for AI in media and entertainment is expected to reach \$99.48 billion by 2030, growing at a CAGR of 26.9% from 2023 to 2030 (Grand View Research, 2023). These statistics underscore the transformative potential of generative AI across various sectors.

In this panel, the panelists will discuss but are not limited to, the following topics.

What will be the impact on academia and industry? Generative AI's influence on academia includes the development of new research methodologies, enhanced educational tools, and collaborative opportunities across disciplines. In industry, it has led to the creation of advanced multimedia applications, personalized content delivery systems, and immersive virtual environments. The ability to generate realistic images, videos, and audio has significant implications for entertainment, marketing, and virtual reality, among other sectors.

What will be the next GenAI products in multimedia? The future of generative AI in multimedia promises the creation of products that are highly interactive and personalized. Examples include:

- (1) Virtual Assistants: These assistants will leverage generative AI to provide in real-time more natural and context-aware interactions.
- (2) Customized Entertainment: AI-generated content tailored to individual preferences, enhancing user experience in gaming, movies, and music.
- (3) Virtual Worlds: Realistic and interactive virtual environments for training, education, and entertainment.

Connections to emerging topics. Beyond the application scenarios of generative AI, we are curious to explore its connection to other emerging techniques such as Embodied AI. Embodied AI, which integrates AI systems into physical entities, is a burgeoning field closely connected to generative AI. The combination of these technologies can lead to the development of sophisticated

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robots and avatars capable of interacting seamlessly with humans in real-world environments. This integration enhances the potential for applications in healthcare, education, and customer service, where human-like interactions are crucial. For example, AI-powered robots in healthcare can assist in patient care, rehabilitation, and surgery, providing support that is both efficient and empathetic.

What are the challenges and opportunities? Generative AI presents both significant challenges and opportunities. **Challenges:** Addressing bias and fairness, ensuring privacy, preventing the

spread of misinformation, and mitigating environmental impact are critical challenges. For instance, generative AI models can inadvertently perpetuate societal biases present in their training data, leading to biased outcomes in their applications. **Opportunities:** Advancing AI-driven content creation, improving human-computer interactions, and developing new applications in various fields offer immense potential for generative AI. The healthcare sector, for example, can benefit from AI-generated diagnostic tools that assist doctors in making more accurate and timely diagnoses.